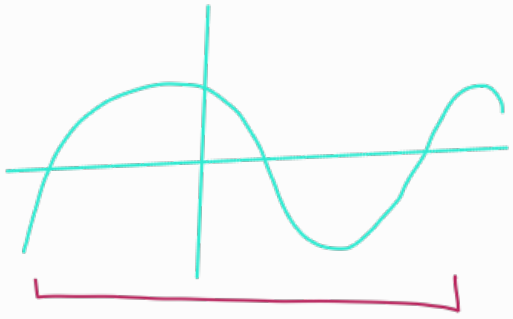
Fourier Series:

1) Periodic ∴ $f(x) = \sin x$, $f(x) = \cos x$

↓

Something that repeats over again at regular intervals.



2) Fourier series only applicable to those functions that are periodic.

$$f(x) = \sum_{n=0}^{\infty} a_n \cos nx + \sum_{n=0}^{\infty} b_n \sin nx$$

Substituting n

=

$$\left[(a_0 \cos 0x) + (a_1 \cos x) + (a_2 \cos 2x) \cdots + (b_0 \sin 0x) + (b_1 \sin x) + (b_2 \sin 2x) \cdots \right]$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos mx + \sum_{n=1}^{\infty} b_n \sin nx$$

actual formula

note:

i) $\sin n\pi = 0$ where n is any number starting from $0, 1, \ldots$

ii) $\cos(\text{even})\pi = 1$

$$\cos 2n\pi = 1$$

where is even

iii) $\cos(\text{odd})\pi = -1$

even.

$$\cos(2n+1)\pi = \cos(2n-1)\pi = -1$$

Integration by parts:

$$\int u\,dv = uv - \int v\,du$$

we need to find

$u =$

$v =$

$du =$

$dv =$

Solve:

$$\int \underbrace{x}_{u} \underbrace{e^x\,dx}_{dv}$$

$u = x \rightarrow du = 1\,dx$

$dv = e^x\,dx$

↳ Integrate this to get v

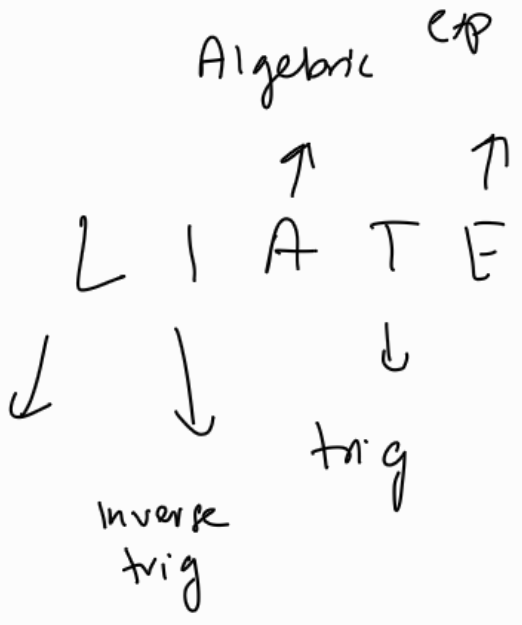
$$v = \int e^x\,dx = e^x$$

Pick your u that when we differentiate it gets more simplified. Reduces.

$$\int x e^x \, dx = x e^x - \int e^x \, dx$$

$$= \boxed{x e^x - e^x}$$

Trick to pick u correctly:



Solve:

$$\boxed{\int x^2 \sin 2x \, dx}$$

Even functions: $f(x) = f(-x)$

e.g $f(x) = x^2$

$f(-x) = (-x)^2 = x^2$

e.g $f(x) = \cos x$

$f(-x) = \cos(-x) = \cos x$

Odd functions:-

$$f(x) \neq f(-x)$$

e.g $f(x) = x^3$

$$f(-x) = (-x)^3 = -x^3$$

e.g $f(x) = \sin x$

$$f(-x) = \sin(-x) = -\sin x$$

1) $$\int e^{ax} \cdot \sin bx \, dx = \frac{e^{ax}}{a^2+b^2}\left(a\sin bx - b\cos bx\right)$$

2) $$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2+b^2}\left(a\cos bx + b\sin bx\right)$$

Types of Question:- | Limits

- type 1: $0 \to 2\pi$
- type 2: $(-\pi$ to $\pi)$ ↓ even/odd
- type 3: $(0$ to $2a)$, a is constant
- type 4: $(-a$ to $a)$

Fourier coefficents → a_0, a_n, b_n

$$a_0 = \frac{1}{2\pi}\int_c^{c+2\pi} f(x)\,dx \qquad a_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\sin nx\,dx$$

Type I limit 0 to 2π $\boxed{c=0}$

1) find fourier series of $f(x) = x^2$ in $(0, 2\pi)$

$$x^2 : \quad a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

finding $a_0 = \frac{1}{2\pi}\int_0^{2\pi} x^2\,dx$

$$a_0 = \frac{1}{2\pi}\left|\frac{x^3}{3}\right|_0^{2\pi} = \frac{1}{2\pi}\left|\frac{(2\pi)^3}{3}\right| - (0)$$

$$= \boxed{\frac{4\pi^2}{3}}$$

finding

$$a_n = \frac{1}{\pi}\int_0^{2\pi} x^2 \cos nx\,dx$$

$x^2 \to u$, $\cos nx \to v$

$\cos \to -\sin$

$\sin$

$$\int uv\,dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 \cdots\cdots$$

$$= \frac{1}{\pi}\left[x^2\left(\frac{\sin(nx)}{n}\right) - 2x\left(\frac{-\cos(nx)}{n^2}\right) + 2\left(\frac{-\sin(nx)}{n^3}\right)\right]_0^{2\pi}$$

$$\boxed{\sin(n\pi) = 0}$$

$$\frac{1}{n^2\pi}\left[-2x(-\cos(nx)) \right]$$

$$\frac{1}{n^2\pi}\left(2x\cos(nx) \right)$$

$$\frac{2}{n^2\pi}\left(x\cos(nx) \right)_0^{2\pi}$$

$$\frac{2}{n^2\pi}\left[2\pi\cos(2\pi n) \right] - \left[0 \right]$$

$$\frac{2}{n^2\pi}\left[2\pi \quad 1 \right] = \frac{4\pi}{n^2\pi} = \boxed{\frac{4}{n^2}}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} x^2 \sin nx\,dx$$

$u \quad v$

$$\int uv\,dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 \cdots\cdots$$

$$= \frac{1}{\pi}\left[x^2\left(\frac{-\cos nx}{n}\right) - 2x\left(\frac{-\sin nx}{n^2}\right) + 2\left(\frac{\cos nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[(2\pi)^2\left(\frac{-\cos n2\pi}{n}\right) + 2\left(\frac{\cos 2n\pi}{n^3}\right)\right] - \left[2\left(\frac{\cos 0}{n^3}\right)\right)$$

$$= \frac{1}{\pi}\left(\frac{-4\pi^2}{n} + \frac{2}{n^2} - \frac{2}{n^2}\right) = \boxed{\frac{-4\pi}{n}}$$

$$x^2 = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}\cos nx + \sum_{n=1}^{\infty}\left(\frac{-4\pi}{n}\right)\sin nx$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n\left(\cos\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n\left(\sin\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{2l}\int_{\text{initial}}^{\text{final}} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$l = \dfrac{\text{final} - \text{initial limit}}{2}$ → formula to find L Value

e.g find the fourier Series $f(x) = \begin{cases} \pi x & 0 < x < 1 \\ \pi(2-x) & 1 < x < 2 \end{cases}$

find value of $L = \frac{2-0}{2} = \boxed{1}$

$$a_0 = \frac{1}{2}\int_0^2 f(x)\,dx$$ = Breaking into two functions.

$$\frac{1}{2}\left[\int_0^1 \pi x\,dx + \int_1^2 \pi(2-x)\,dx\right]$$

$$\frac{1}{2}\left[\left(\frac{\pi x^2}{2}\right)_0^1 + \left[\frac{\pi(2-x)^2}{-2}\right]_1^2\right]$$

$$\frac{\pi}{2}\left(\frac{x^2}{2}\right)_0^1 + \frac{\pi}{2}\left[\right.$$